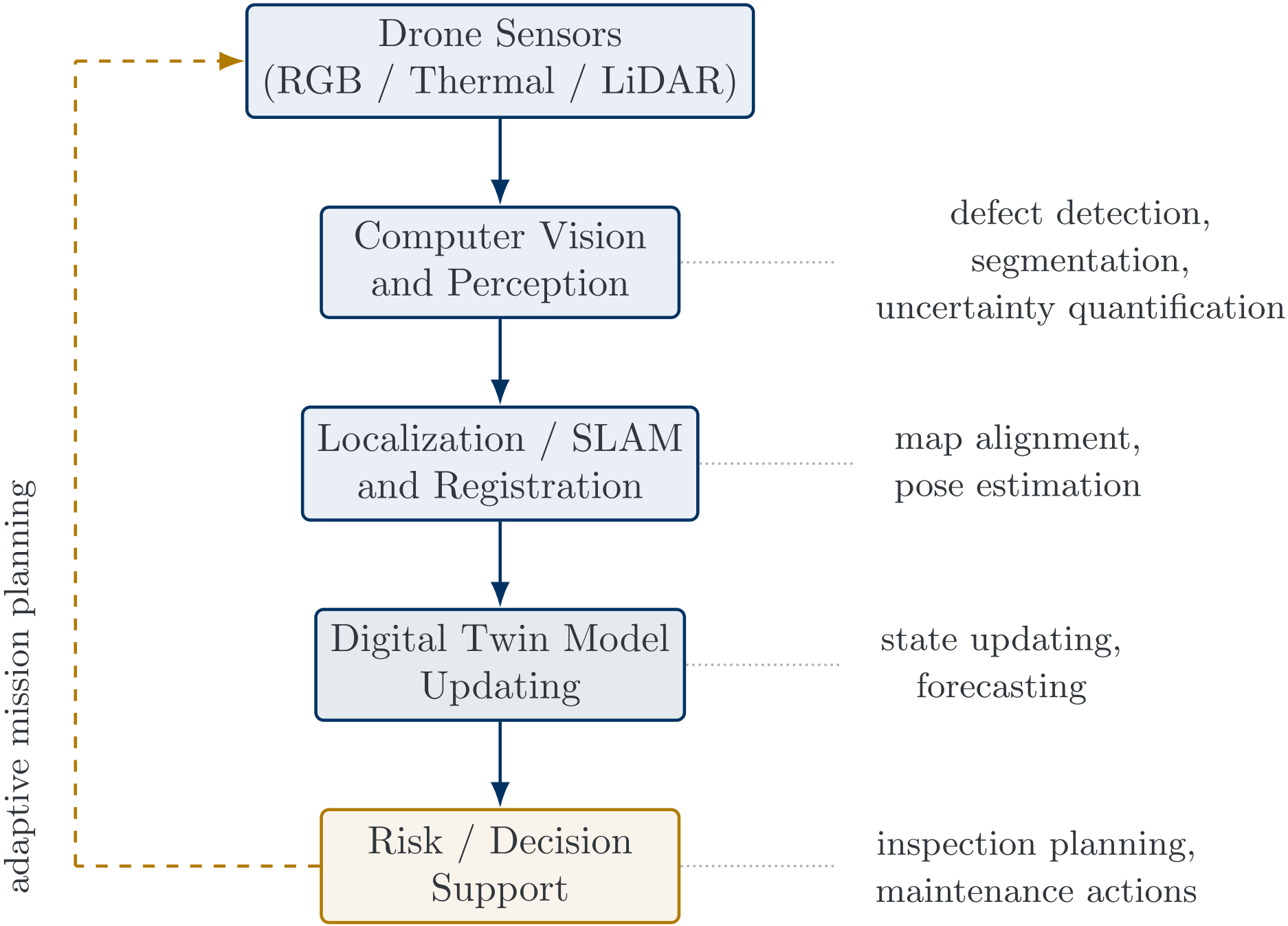
Drone Sensors
(RGB / Thermal / LiDAR)
Computer Vision
and Perception
defect detection,
segmentation,
uncertainty quantification
Localization / SLAM
and Registration
map alignment,
pose estimation
Digital Twin Model
Updating
state updating,
forecasting
Risk / Decision
Support
inspection planning,
maintenance actions
adaptive mission planning